WORKED SOLUTION

STEP I 2018, Question 9

Mechanics • Motion on two inclined roads

STEP I 2018, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

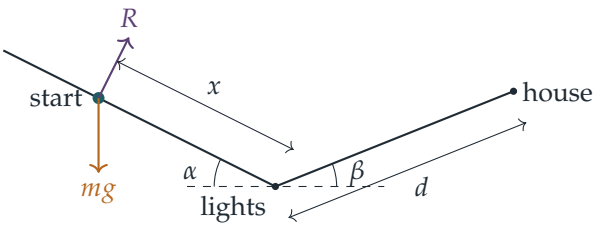
step.maths.org/questions/18-s1-q9

The key idea

With no resistance, the go-kart has constant acceleration on each straight section: $g\sin\alpha$ down the first slope and a deceleration of $g\sin\beta$ up the second. So each section is a constant-acceleration problem, and the two are glued together by the speed at the traffic lights. Energy (or $v^2 = u^2 + 2as$) settles whether I get home at all; the timing comes from $s = ut + \frac{1}{2}at^2$ on each section, and the rest is careful algebra in terms of k.

Solution

Model the go-kart and rider as a particle moving along each straight line of greatest slope, with no friction or air resistance. The only forces are the weight mg and the normal reaction R, which does no work.



Reaching the house

Resolving along the first slope, the component of the weight down the slope is $mg\sin\alpha$, so the acceleration is $g\sin\alpha$. Starting from rest and travelling a distance x, the speed V at the lights satisfies

$$V^2 = 2gx\sin\alpha.$$

Since there is no sudden change of speed, I start up the second section with speed V, and now the weight gives a deceleration $g\sin\beta$ along the slope. At distance s beyond the lights my speed v satisfies $v^2 = V^2 - 2gs\sin\beta$. To reach the house, v^2 must stay

non-negative all the way to $s = d$, i.e.

$$2gx\sin\alpha - 2gd\sin\beta \geq 0 \quad \Longrightarrow \quad x\sin\alpha \geq d\sin\beta,$$

as required. (Equivalently: by conservation of energy I can never rise above my starting height, which is $x\sin\alpha$ above the lights, while the house is $d\sin\beta$ above them.)

The total time

To keep the algebra tidy, write $c = \left(\frac{1}{2}g\sin\alpha\right)^{1/2}$, so $g\sin\alpha = 2c^2$ and, since $\sin\beta = \sin\alpha/k$, $g\sin\beta = 2c^2/k$.

First section. From $x = \frac{1}{2}(g\sin\alpha)t_1^2 = c^2t_1^2$ the time to the lights is $t_1 = \sqrt{x}/c$, and the speed there is $V = (g\sin\alpha)t_1 = 2c\sqrt{x}$.

Second section. Measuring s from the lights, $s = Vt - \frac{1}{2}(g\sin\beta)t^2$, so the time t_2 to reach $s = d$ satisfies

$$\frac{c^2}{k}t_2^2 - 2c\sqrt{x}\,t_2 + d = 0.$$

? **Ask yourself** This quadratic has two roots. Which one is the time I arrive?

The larger root is the time at which a particle on an endlessly long slope would pass $s = d$ again on the way back down, after stopping. I arrive the first time I reach $s = d$, so we want the smaller root:

$$t_2 = \frac{2c\sqrt{x} - \sqrt{4c^2x - 4c^2d/k}}{2c^2/k} = \frac{k}{c}\left(\sqrt{x} - \sqrt{x - \tfrac{d}{k}}\right) = \frac{1}{c}\left(k\sqrt{x} - \sqrt{k^2x - kd}\right),$$

using $k\sqrt{x - d/k} = \sqrt{k^2x - kd}$ as $k > 0$. The square root is real precisely because $x \geq d/k$, which is the condition $x\sin\alpha \geq d\sin\beta$ from the first part.

Adding, $T = t_1 + t_2$ gives

$$\left(\frac{g\sin\alpha}{2}\right)^{1/2} T = cT = (1+k)\sqrt{x} - \sqrt{k^2x - kd},$$

which is the required result.

Minimising T

Since c is a positive constant, we minimise $f(x) = (1+k)\sqrt{x} - \sqrt{k^2x - kd}$ over $x \geq d/k$. Differentiating,

$$f'(x) = \frac{1+k}{2\sqrt{x}} - \frac{k^2}{2\sqrt{k^2x - kd}}.$$

Setting $f'(x) = 0$ gives $(1+k)\sqrt{k^2x - kd} = k^2\sqrt{x}$. Both sides are positive, so squaring loses nothing:

$$\begin{aligned}(1+k)^2(k^2x - kd) &= k^4x\\ x\left[(1+k)^2k^2 - k^4\right] &= (1+k)^2kd\\ xk^2(1+2k) &= (1+k)^2kd,\end{aligned}$$

using $(1+k)^2 - k^2 = 1 + 2k$. Dividing by $k^2(1+2k) > 0$,

$$x = \frac{(1+k)^2}{k(1+2k)}\,d.$$

This is an allowed starting point: $(1+k)^2 = 1 + 2k + k^2 > 1 + 2k$, so this x is strictly greater than d/k and I do reach the house.

It makes sense that there is a best x: start too close and I crawl over the last stretch of the hill; start too far and I spend a long time on the first slope.

Checks

- At the optimum, $\sqrt{k^2x - kd} = k^2\sqrt{x}/(1+k)$, so $cT = \sqrt{x}\,\dfrac{1+2k}{1+k} = \sqrt{\dfrac{(1+2k)d}{k}}$, a pleasingly simple minimum time.
- Take $\alpha = \beta$, so $k = 1$. Then the optimum is $x = \frac{4}{3}d$, and the formula gives $cT = 2\sqrt{4d/3} - \sqrt{d/3} = \sqrt{3d}$, agreeing with the previous check.
- At the threshold $x = d/k$ I just creep to the door, $f'(x) \to -\infty$, so moving the start back a little certainly saves time.
- Stepping the motion forward numerically in small time intervals, for several random angles and distances, reproduced the formula for T to five significant figures; a computer search over x located the minimum at the value found above.

Common slips

- Taking the larger root of the quadratic for t_2: that is the time of coming back down past the house.
- Forgetting that the speed at the lights is not given and must be carried over from the first section.
- Dropping the condition $x \geq d/k$, which is exactly what makes $\sqrt{k^2x - kd}$ real.

Beyond STEP

The tidy answer for the minimum time hints at a broader theme: choosing a path or starting point to minimise travel time is the subject of the *calculus of variations*, whose

most famous problem, the brachistochrone, asks for the curve of fastest descent under gravity. In first-year mechanics you'll also meet the idea that, for conservative forces, energy alone tells you where a particle can go, while time requires integrating the equation of motion. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas carefully.

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